

Yongxue CHEN

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RESEARCH PROFILE

PhD candidate in Production and Manufacturing Engineering, specializing in robotics and advanced manufacturing with a focus on motion planning, trajectory optimization, and process planning.

My research centers on robotic manufacturing and hybrid additive-subtractive processes, particularly for multi-axis robotic systems and complex geometries.

I develop scalable computational methods that integrate geometric modeling, numerical optimization, and learning-based techniques to improve manufacturability, efficiency, and reliability.

EDUCATION

The University of Manchester (UoM), Manchester, UK Sept 2023 – Feb 2027 (Expected)

PhD in Production and Manufacturing Engineering

Supervisor: Prof. Charlie C. L. Wang

Research focus: Robot-assisted manufacturing, hybrid additive-subtractive manufacturing, multi-axis 3D printing, carbon-fiber printing, and process planning

Shanghai Jiao Tong University (SJTU), Shanghai, China

Sept 2020 – Mar 2023

MEng in Mechanical Engineering

Supervisor: Prof. Ye Ding

Research focus: Robot-assisted milling, motion and trajectory planning, and geometric modeling

Shanghai Jiao Tong University (SJTU), Shanghai, China

Sept 2016 – Jun 2020

BEng in Mechanical Engineering

GPA: 3.797/4.3 (Ranked 6/146)

PUBLICATIONS

First-Author Journal Articles

1. **Y. Chen**, T. Liu, Y. Huang, W. Wang, T. Zhang, K. Qian, Z. Shi, and C. C. L. Wang. “Can Any Model Be Fabricated? Inverse Operation Based Planning for Hybrid Additive-Subtractive Manufacturing.” *ACM Transactions on Graphics (SIGGRAPH Asia)*, 2025. (Oral Presentation)
2. **Y. Chen**, T. Zhang, Y. Huang, T. Liu, and C. C. L. Wang. “Co-Optimization of Tool Orientations, Kinematic Redundancy, and Waypoint Timing for Robot-Assisted Manufacturing.” *IEEE Transactions on Automation Science and Engineering*, 2025.
3. **Y. Chen**, Y. Lu, and Y. Ding. “Toolpath Generation for Robotic Flank Milling via Smoothness and Stiffness Optimization.” *Robotics and Computer-Integrated Manufacturing*, 2023.
4. **Y. Chen** and Y. Ding. “Posture Optimization in Robotic Flat-End Milling Based on Sequential Quadratic Programming.” *ASME Journal of Manufacturing Science and Engineering*, 2023.
5. **Y. Chen**, P. Huang, and Y. Ding. “An analytical method for corner smoothing of five-axis linear paths using conformal geometric algebra.” *Computer-Aided Design*, 2022.
6. **Y. Chen**, W. Dong, and Y. Ding. “An efficient method for collision-free and jerk-constrained trajectory generation with sparse way-points for a flying robot.” *Science China Technological Sciences*, 2021.

Co-Author Journal Articles

1. T. Liu, A. Lyu, **Y. Chen**, Y. Jiang, M. J. Petty, and C. C. L. Wang. “Co-Optimization of Structure and Manufacturable Semi-Continuous Layers for Laminated Composites.” *ACM Transactions on Graphics*, 2026.
2. N. Dutta, T. Zhang, T. Liu, **Y. Chen**, and C. C. L. Wang. “Implicit Neural Field-Based Process Planning for Multi-Axis Manufacturing: Direct Control over Collision Avoidance and Toolpath Geometry.” *Computer-Aided Design*, 2026.
3. Y. Huang, R. Su, K. Qian, T. Zhang, **Y. Chen**, T. Liu, G. Fang, W. Wang, and C. C. L. Wang. “Force-based adaptive deposition in multi-axis additive manufacturing: Low porosity for enhanced strength.” *Robotics and Computer-Integrated Manufacturing*, 2026.

4. T. Liu, T. Zhang, **Y. Chen**, W. Wang, Y. Jiang, Y. Huang, and C. C. L. Wang. “Neural Co-Optimization of Structural Topology, Manufacturable Layers, and Path Orientations for Fiber-Reinforced Composites.” *ACM Transactions on Graphics*, 2025.
5. T. Zhang, T. Liu, N. Dutta, **Y. Chen**, R. Su, Z. Zhang, W. Wang, and C. C. L. Wang. “Toolpath generation for high density spatial fiber printing guided by principal stresses.” *Composites Part B*, 2025.
6. T. Liu, T. Zhang, **Y. Chen**, W. Wang, Y. Huang, and C. C. L. Wang. “Neural Slicer for Multi-Axis 3D Printing.” *ACM Transactions on Graphics*, 2024.
7. Y. Ding and **Y. Chen**. “Point Pair-Based Expression of Cutter Swept Envelopes in Five-Axis Milling.” *Computer-Aided Design*, 2023.

CONFERENCE PRESENTATIONS

1. **ACM SIGGRAPH Asia**, Hong Kong Dec 2025
Oral Presentation: “Can Any Model Be Fabricated? Inverse Operation Based Planning for Hybrid Additive-Subtractive Manufacturing”
2. **IEEE/RSJ IROS**, Hangzhou, China Oct 2025
Oral Presentation: “Co-Optimization of Tool Orientations, Kinematic Redundancy, and Waypoint Timing for Robot-Assisted Manufacturing”

PATENTS

1. **Y. Chen**, W. Dong, and Y. Ding. “Unmanned Aerial Vehicle Speed Planning Method.” Granted Chinese Invention Patent, CN110879614B.
2. Y. Ding, **Y. Chen**, and W. Dong. “Collision-Free Path Planning Method for Unmanned Aerial Vehicles.” Granted Chinese Invention Patent, CN112504279B.
3. Y. Ding and **Y. Chen**. “Pose Optimization Method for Flat-End Milling with Industrial Robots.” Granted Chinese Invention Patent, CN115016258B.
4. Y. Ding and **Y. Chen**. “Trajectory Smoothing Method for Linear Motion of a Robotic Arm.” Granted Chinese Invention Patent, CN114536337B.
5. Y. Ding, **Y. Chen**, and Q. Li. “Analytical Calculation Method for the Envelope Surface of Rotary Cutting Tools.” Granted Chinese Invention Patent, CN112487657B.

RESEARCH EXPERIENCE

- Neural Field Optimization for Hybrid Manufacturing** UoM | May 2025 – May 2026
- Developed a neural field-based optimization framework for hybrid additive-subtractive manufacturing, jointly optimizing fabrication sequences and tool orientations for complex geometries.
 - Formulated an FEA-guided, distortion-aware optimization strategy to improve the mechanical stability of intermediate structures while considering manufacturability constraints.
 - Improved efficiency on large-scale models, produced more continuous fabrication paths, and enhanced manufacturing accuracy; first-author journal manuscript currently under review.
- Inverse Operation-Based Planning for Hybrid Manufacturing** UoM | May 2024 – May 2025
- Developed an inverse-operation-based planning framework for hybrid additive-subtractive manufacturing, formulating fabrication as a reverse process from the target geometry to an initial state.
 - Formulated localized stability constraints and an efficient planning strategy to ensure scalability and reduce redundant operations in complex fabrication sequences.
 - Enabled the fabrication of complex geometries without restrictive manufacturability constraints, demonstrating a new paradigm for hybrid manufacturing; published in *ACM Transactions on Graphics (SIGGRAPH Asia 2025)*, with oral presentation.
- Trajectory Co-Optimization for Robot-Assisted Manufacturing** UoM | Sept 2023 – May 2024
- Developed a unified framework jointly optimizing tool orientations, kinematic redundancy, and waypoint timing for multi-axis robot-assisted manufacturing.
 - Formulated the problem as a large-scale constrained optimization over kinematic redundancy and temporal parameters, and introduced a decomposition-based scheme for efficient solution.
 - Achieved smooth and efficient robot motions across complex and large-scale toolpaths; results published in *IEEE Transactions on Automation Science and Engineering*, 2025.

	6-DoF Path Optimization in Robotic Flank Milling SJTU Oct 2022 – Mar 2023 - Developed a unified optimization framework for 6-DoF toolpath planning in robotic flank milling, jointly considering machining accuracy, kinematic smoothness and robot stiffness. - Derived analytical models using conformal geometric algebra to evaluate machining errors and stiffness, enabling efficient and accurate gradient-based optimization. - Improved machining quality and computational efficiency compared to decoupled approaches; results published in <i>Robotics and Computer-Integrated Manufacturing</i>, 2023.
	Posture Optimization in Robotic Flat-End Milling SJTU Oct 2021 – Oct 2022 - Developed a sequential quadratic programming-based optimization framework for jointly optimizing tool orientation and kinematic redundancy under geometric and kinematic constraints. - Formulated machining objectives and constraints to enable efficient and stable optimization of tool motion while avoiding geometric interference. - Established a foundation for subsequent work on unified toolpath optimization in robotic manufacturing; results published in <i>ASME Journal of Manufacturing Science and Engineering</i>, 2023.
	Corner Smoothing of Five-Axis Toolpaths SJTU Mar 2021 – Oct 2021 - Built a unified analytical conformal-geometric-algebra framework for five-axis toolpath smoothing, constructing G^3-continuous transitions in 6D without iterative synchronization. - Improved motion smoothness and efficiency; published in <i>Computer-Aided Design</i>, 2022.
	Analytical Modeling of Cutter Swept Envelopes SJTU Sept 2020 – Feb 2023 - Developed a compact, unified model of cutter swept envelopes in multi-axis machining using conformal geometric algebra and point-pair primitives, without local coordinate transformations. - Improved robustness and efficiency; published in <i>Computer-Aided Design</i>, 2023.
	Trajectory Planning for Quadrotors SJTU Jul 2019 – Sept 2020 - Generated smooth, collision-free, jerk-constrained trajectories via sequential convex programming. - Outperformed prior methods in computation time; published in <i>Sci. China Technol. Sci.</i>, 2021.
TEACHING EXPERIENCE	Teaching Assistant , The University of Manchester 2026 Additive Manufacturing & 3D Product Modelling Taught 3D scanning/printing labs and assessed reports with design/manufacturing feedback.
	Teaching Assistant , Shanghai Jiao Tong University 2021 MATLAB and Its Applications in Engineering Designed and graded coursework assignments, and supported students through Q&A.
ACADEMIC SERVICE	Reviewer for international journals and conferences, including: ACM SIGGRAPH Asia, <i>IEEE Transactions on Visualization and Computer Graphics</i> , <i>IEEE/ASME Transactions on Mechatronics, Robotics and Computer-Integrated Manufacturing</i> , <i>Computer-Aided Design</i> , <i>Journal of Manufacturing Processes</i> .
SELECTED HONORS	Departmental Doctoral Scholarship, The University of Manchester 2023
	Shanghai Municipal Outstanding Graduate 2023
	Outstanding Student Award, Shanghai Jiao Tong University 2022
	National Scholarship for Postgraduate Students (China) 2021
	Outstanding Graduate, Shanghai Jiao Tong University 2020
TECHNICAL EXPERTISE	Research Areas: Robotics, advanced manufacturing, computational process planning Methods: Learning-based methods, geometric modeling, numerical optimization, geometric algebra Programming: C++, Python, MATLAB